

FieldAI - Product Requirements Document (PRD)

Enterprise Multi-Agent Audit Fieldwork & Process Intelligence Platform

Document ID:	PRD-FIELDAI-2026-V1.0	Product Stage:	Production / Active Deployment
System Version:	v1.0 (Full Multi-Agent Architecture)	Target Deployment:	Render Cloud (Docker + PostgreSQL)
Prepared By:	Sami Associates Internal Audit Tech Practice	Product Owner:	Sami (Leadership Review Approved)
Core Technologies:	LangGraph Streamlit PostgreSQL ElevenLabs DeepSeek	Date of Specification:	October 2026

Executive Summary: FieldAI is an enterprise-grade, multi-agent AI audit fieldwork platform engineered to automate the transformation of unstructured audit walkthrough interviews, meeting recordings, and auditee documents (SOPs, policies, manuals) into a linked, rigorous, and verifiable internal audit workpaper package. The system replaces manual transcription, narrative drafting, flowchart drawing, and RCM generation with a coordinated orchestration of 21 specialized LangGraph agents, an interactive 15-page Streamlit portal, a reliable PostgreSQL state store and job queue, and a dedicated asynchronous background worker. Every single workpaper element (step, risk, control, test, finding) maintains bi-directional source traceability with second-level audio timestamps and document clause citations, ensuring compliance with the IIA Global Internal Audit Standards (2024).

1. System Architecture & Infrastructure Topology

FieldAI adopts a decoupled, multi-tier microservices architecture defined declaratively via a Render Blueprint (render.yaml). The system coordinates web presentation, background computing, recurring cron testing, database persistence, and external AI providers.

Tier / Service	Component	Technology Stack	Operational Role & SLA
Frontend / Web	fieldai-web	Streamlit 1.35+, Python 3.11, Docker	Interactive UI for auditors, 15 module pages, data editors, Cytoscape/Graphviz viewers, live audio input.
Asynchronous Worker	fieldai-worker	worker.py, Python 3.11, Docker	Picks up long-running tasks (transcription, OCR, analytics, process mining) via PostgreSQL SKIP LOCKED queue.
Recurring Scheduler	fieldai-monitoring	jobs/run_monitoring.py, Cron	Executes weekly continuous monitoring scripts, evaluates recurring tests against latest data extracts.
Relational Store	fieldai-db	PostgreSQL 16 (psycopg 3 pool)	Single source of truth: master process models, versions, audit trail, LangGraph checkpoints, and binary files.
Speech-to-Text	STT Engine	ElevenLabs Scribe / OpenAI gpt-4o	Primary diarization and transcription with word-level timestamps in Arabic, English, or mixed speech.
LLM Reasoning	Cognitive Core	DeepSeek Flash (OpenAI SDK compat)	Process extraction, risk/control mapping, 5-category reconciliation, test procedures, and QA checks.

2. Comprehensive Application Screen Catalogue (15 Modules)

The web user interface is partitioned into 15 purpose-built Streamlit modules under pages/, designed specifically for audit teams to progress systematically from engagement setup to fieldwork delivery.

Page & File	Module Name	Key Functional Capabilities & User Interactions
app.py	Portal & Auth	Bcrypt login authentication, role-based page navigation, active engagement selector, global Ask FieldAI copilot.
1_Engagements.py	Engagements & Scoping	Create/manage audit engagements, define process entities, assign stable code prefixes (e.g. P2P, O2C, HTR), track completion.
2_Capture.py	Walkthrough Capture	Browser in-person recording (st.audio_input), multi-format audio/video upload, mandatory consent logging, diarization, redaction, near-live co-pilot.
3_Review_Changes.py	Change Set Review	Multi-meeting diff analysis; tracked changes review (Added/Changed/Confirmed/Contradicted/Removed); side-by-side conflict resolution; version bump.
4_Process_Table.py	Master Process Table	Interactive tabular view of steps, owners, systems, risks, controls; in-place data editor; source citation links; confidence ratings.

5_Flowchart.py	Interactive Flowcharts	Dual-engine rendering: Cytoscape.js interactive canvas (Dagre layout, Bezier curves, draggable nodes, density presets) and Graphviz Blueprint (1:1 scale). 5 export formats.
6_RCM.py	Risk-Control Matrix	Inherent risk evaluation, control design adequacy ratings, AI suggestions for unmitigated gaps, framework mapping (COSO, COBIT, ISO, NCA, SAMA).
7_Audit_Program.py	Audit Test Program	Generates ToD and ToE steps; deterministic sample size calculation based on frequency and risk; PBC request lists; manager approval and workpaper lock.
8_Documents.py	Document Reconciliation	Document register; multi-format upload (PDF OCR, DOCX, XLSX); 5-category reconciliation against walkthrough statements; document citation Q&A.
9_Prep.py	Meeting Preparation	Generates bilingual (Arabic/English) walkthrough question packs, follow-up interview agendas, and risk-based scoping matrices.
10_Testing.py	Field Testing & Analytics	5 testing engines: Automated Analytics (duplicate payments, split POs, Benford's Law), Segregation of Duties (SoD), Process Mining, Evidence OCR, Recurring Tests.
11_Findings.py	Findings & QA	Drafts 5 Cs findings (Condition, Criteria, Cause, Effect, Recommendation); automated QA review against IIA Global Standards 2024 checklist.
12_Dashboard.py	Executive Dashboard	CAE executive metrics, process risk heatmaps, control gap analytics, fieldwork completion tracking, automated exception alerts.
13_Confirm.py	Auditee Confirmation	Secure token-based external access portal for auditees to review process narratives, inspect visual swimlanes, and submit validation feedback.
14_Admin.py	System Administration	User management and RBAC, sampling parameter tables, risk and control reference libraries, AI model credentials, immutable audit log viewer.
15_Centralized_Storage.py	Centralized Storage	Enterprise document repository: hierarchical folder tree, multi-tag filtering, metadata indexing, document preview, secure downloads, storage governance.

3. Multi-Agent Ecosystem Specifications (21 Agents)

FieldAI executes all substantive intelligence tasks through 21 autonomous, single-responsibility agents located in agents/. Agents are orchestrated as nodes in LangGraph state graphs with typed state contracts (FieldAIState).

#	Agent Module	Engine	Operational Function & Core Logic	Key Outputs
1	transcription_agent	ElevenLabs Scribe	Transcribes audio/video with speaker diarization, word-level timestamps, and language detection. Redaction support.	transcript_segments
2	summary_agent	DeepSeek Flash	Generates concise executive summaries, open questions, and preliminary PBC request lists.	summary, open_items, pbc
	process_extraction_agent	DeepSeek Flash	Extracts ordered steps, responsible roles, departments, systems, decision points, and handoffs from text.	ProcessStep objects
3	risk_control_agent	DeepSeek Flash	Identifies stated and library-suggested risks and controls, categorizes nature/frequency/owner, flags SoD gaps.	Risk & Control objects
4	change_set_agent	DeepSeek Flash	Compares new extractions with master model. Resolves entities, generates tracked changes, pauses for auditor review.	change_set, conflict_list
5	document_agent	DeepSeek + OCR	Parses policies, SOPs, and manuals (PDF, DOCX, XLSX). Registers metadata and chunks text with section citations.	doc_chunks, doc_metadata
6	reconciliation_agent	DeepSeek Flash	Performs 5-category gap analysis comparing documented procedures with walkthrough interview statements.	recon_items, gap_notes
7	flowchart_agent	Deterministic	Generates Graphviz DOT and Cytoscape JSON models with swimlane clustering, risk/control badges, and density modes.	dot_code, cy_elements
8	rcm_agent	DeepSeek Flash	Constructs Risk-Control Matrix, evaluates design adequacy, suggests gap controls, and maps regulatory frameworks.	rcm_rows, design_ratings
9	audit_program_agent	DeepSeek + Rules	Drafts ToD and ToE procedures, computes deterministic sample sizes from sampling table, defines PBC requirements.	test_steps, sample_basis
10				
11	prep_agent	DeepSeek Flash	Generates bilingual walkthrough questionnaires, scoping matrices, and tailored follow-up interview agendas.	question_pack, scope_plan
12	copilot_agent	Scribe + DeepSeek	Near-live interview assistant analyzing audio chunks to suggest real-time probing questions and capture PBC items.	live_prompts, instant_pbc
13	knowledge_agent	DeepSeek Flash	Enables cross-engagement intelligence reuse, baseline process templates, and year-on-year change detection.	baseline_model, yoy_diff
14	doc_qa_agent	DeepSeek Flash	Performs semantic search across transcripts and documents, answering auditor queries with exact citations.	cited_answers
15	analytics_agent	Pandas + Rules	Maps data schemas and executes fixed audit analytics (duplicate payments, split purchases, Benford's Law).	test_results, exceptions
16	evidence_agent	OCR + DeepSeek	Inspects sample evidence files (PO, invoice, approval email) and verifies compliance with audit test attributes.	attribute_results, quotes
17	sod_agent	Rules + DeepSeek	Evaluates user-access matrices against toxic role combination rules, mapping conflicts to swimlane handoffs.	sod_conflicts, role_gaps
18	process_mining_agent	Pandas + DeepSeek	Analyzes event logs to identify process variants, calculate throughput times, and detect control bypasses.	variants, bypass_alerts
19	monitoring_agent	Deterministic	Schedules recurring automated tests, logs periodic execution runs, and triggers exception alerts.	cron_logs, new_exceptions
20	findings_agent	DeepSeek Flash	Drafts formal audit findings in the 5 Cs format (Condition, Criteria, Cause, Effect, Recommendation).	Finding objects, ratings
	qa_review_agent	DeepSeek Flash	Reviews workpapers against IIA Global Standards 2024 and methodology checklist, generating review notes.	qa_checklist, review_notes
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4. Visualization & Deliverables Engine

FieldAI provides a specialized dual-engine visualization architecture designed to handle complex audit flowcharts with non-bold Inter typography, curved connectors, swimlane segregation, and high-density information compacting.

4.1 Cytoscape.js Interactive Canvas (Dagre + Bezier Routing):

- Layout Engine: Dagre hierarchical DAG layout with top-to-bottom orientation (rankDir: 'TB'), tight node/rank separation (nodeSep: 35, rankSep: 45), and automatic parent-compound bounding for swimlanes.
- Connector Styling: Smooth curved Bezier curves (curve-style: 'bezier') with obstacle-aware routing to eliminate jagged angles.
- Audit Badges: Inherent risk pills (crimson R-xx), control badges (emerald C-xx), decision diamonds (amber), and system cylinders.

- Density Presets: Supports three layout compaction modes: Micro (ultra-dense), Compact (default 8pt font, tight padding), and Full.
- Interactivity: Fully draggable nodes, click-to-inspect audit metadata side drawer, zoom controls (+In, -Out, Fit, Reset), PNG and JSON exports.

4.2 High-Fidelity Graphviz Architecture Blueprint:

- Scale & Routing: Rendered at authentic 1:1 pixel scale (use_container_width=False) preventing oversized magnification. Smooth spline routing (splines=spline) with left-aligned swimlane cluster labels.
- Geometry: Compact pill shapes for Start/End terminals, proportioned decision diamonds, word-wrapped process text, and eliminated overhead header boxes.

4.3 Multi-Format Export Deliverables:

Export Format	Extension	Target Application & Interoperability
BPMN 2.0 XML	.bpmn	Standard OMG BPMN 2.0 with full swimlanes, gateway nodes, and sequence flows for Camunda, Signavio, and enterprise BPM suites.
draw.io / Diagrams	.drawio	Native draw.io XML with swimlane containers and connected vertices. Can be directly converted to Microsoft Visio (.vsdx).
Audit Summary PDF	.pdf	Publication-quality formal deliverable generated via ReportLab with Inter typography, step tables, and risk/control listings.
Graphviz DOT	.dot	Raw Graphviz source code formatted with cluster swimlanes for automated CI/CD documentation pipelines.
Cytoscape JSON	.json	Complete node/edge element graph and layout metadata for web-based graph visualization engines.

5. Data Architecture, Schema & Relational Models

FieldAI persists all operational state in PostgreSQL (or local SQLite fallback) via `core/db.py` and `core/storage.py`. The relational schema supports complete audit trails, stable entity coding, version snapshots, and LangGraph checkpoints.

Table Name	Primary Keys & Foreign Keys	Key Columns & Attributes	Functional Purpose & Retention
users	id (PK)	username, password_hash, role, active, created_at	User identity and RBAC (auditor, manager, admin, auditee).
engagements	id (PK)	name, entity, period, status, created_at	Top-level audit engagement container (e.g. FY26 Internal Audit).
processes	id (PK), engagement_id (FK)	name, code_prefix, current_version, status	Audit process definition (e.g. P2P, Treasury) with prefix.
sources	id (PK), process_id (FK)	kind, title, storage_path, language, consent_recorded	Raw evidence input registry (audio, video, document, SOP).
transcript_segments	id (PK), source_id (FK)	speaker_label, speaker_name, start_s, end_s, text, redacted	Diarized transcript utterances with second-level timestamps.
steps	id (PK), process_id (FK)	step_code, order_num, description, role, system, is_decision	Master process flow steps. Never renumbered; status='withdrawn'.
risks	id (PK), process_id (FK)	risk_code, description, inherent_rating, ai_suggested	Identified inherent risk catalogue with severity scoring.
controls	id (PK), process_id (FK)	control_code, description, type, nature, freq, key_control	Internal controls with design adequacy ratings and framework refs.
versions	id (PK), process_id (FK)	version_label, snapshot_json, change_log, approved_by	Immutable version snapshots (v1.0, v1.1) approved by managers.
change_sets	id (PK), process_id (FK)	source_id, status (proposed/applied), changes_json	Multi-meeting delta proposals awaiting human auditor review.
documents	id (PK), source_id (FK)	doc_type, owner, version, effective_date, approval_status	Auditee document metadata register with freshness tracking.
recon_items	id (PK), process_id (FK)	category, item_code, doc_ref, meeting_ref, note, status	5-category reconciliation items between SOPs and interviews.
tests	id (PK), control_id (FK)	test_code, objective, test_type, procedure, sample_size	Audit program test steps linked to deterministic sampling rules.
test_runs	id (PK), test_id (FK)	run_type, parameters_json, result_summary, exceptions_json	Substantive test executions and continuous monitoring log.
findings	id (PK), process_id (FK)	finding_code, condition, criteria, cause, effect, recommendation	5 Cs formal audit observations linked to controls and tests.
files	id (PK)	filename, mime_type, size_bytes, data (BYTEA), sha256	Centralized binary storage accessible to web and worker.
jobs	id (PK)	job_type, payload_json, status, progress, message, thread_id	Asynchronous worker task queue (SELECT ... FOR UPDATE SKIP LOCKED).
audit_log	id (PK)	user_id, action, entity, entity_id, details_json, created_at	Append-only immutable audit trail recording all user actions.

6. Non-Functional Requirements, Security & Standards

FieldAI adheres to strict security, auditability, and internal audit regulatory standards:

- IIA Standards Compliance: Fully aligned with the IIA Global Internal Audit Standards (2024 Edition), ensuring complete documentation of planning, walkthrough narratives, risk-control matrices, substantive test procedures, and 5 Cs findings.
- Bi-Directional Traceability: 100% of generated steps, risks, controls, tests, and findings maintain clickable or referenced links to raw evidence (audio timestamp e.g. 00:14:22 or document section e.g. SOP v2.1 Section 4.2).
- Zero AI Code Execution: The system strictly prohibits executing arbitrary LLM-generated code against client financial data. All substantive testing relies exclusively on verified, deterministically coded Python/Pandas functions in `analytics/library.py`.
- Auditor-in-the-Loop Safeguards: All AI extractions are treated as draft recommendations. Human auditors retain full override authority. Changes to key controls require explicit managerial re-approval before updating master baselines.
- Data Privacy & Redaction: Meeting recordings require recorded participant consent. Audio segment redaction allows auditors to expunge sensitive personal data or off-the-record conversations prior to LLM processing.
- Immutable Audit Trail: Every user login, model modification, change-set approval, and export action is permanently logged to the append-only `audit_log` table with SHA-256 integrity verification.

7. Acceptance Criteria & Production Verification

- [PASS] Browser audio recording and multi-format audio/video uploads transcribe cleanly with speaker diarization and second timestamps.
- [PASS] Process tables accurately reflect extracted steps, owners, systems, and decision branches with associated confidence ratings.
- [PASS] Interactive Cytoscape flowcharts render with smooth Bezier curves, draggable nodes, and click-to-inspect audit metadata drawers.
- [PASS] Graphviz architecture blueprints render at compact 1:1 scale without oversized font or shape distortion.
- [PASS] All 5 deliverable formats (BPMN 2.0 XML, draw.io XML, Audit Summary PDF, Graphviz DOT, Cytoscape JSON) export without runtime errors.
- [PASS] Follow-up meetings generate structured change sets classifying items as Added, Changed, Confirmed, Contradicted, or Removed.
- [PASS] Uploaded SOPs generate 5-category reconciliation reports comparing documented procedures against interview statements.
- [PASS] Substantive analytics, SoD checks, and process mining engines execute deterministically on uploaded client data files.
- [PASS] All automated test suites (including tests/test_flowchart.py) pass with 100% success rate in CI/CD pipeline.

Prepared by: Lead Audit Architect

Signature: _____

Date: October 2026

Approved by: Product Owner (Sami)

Signature: _____

Status: Approved for Production Deployment